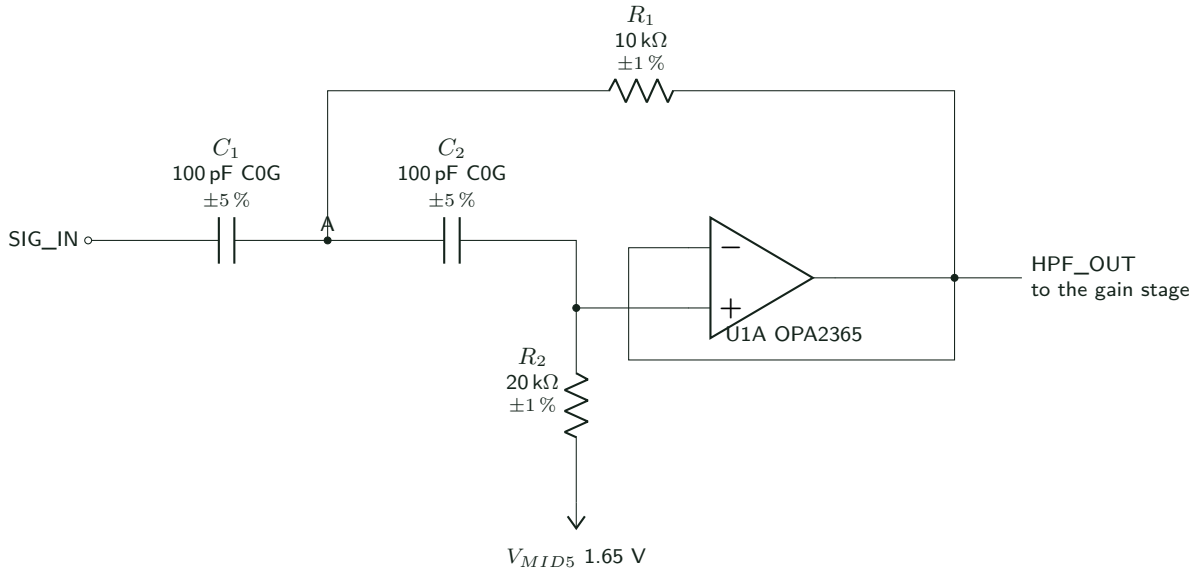
Slide 4: high-pass filter

Cut the plant noise below the AE band

ADS7057 test board v3 — study notes

1 Why the board needs a high-pass

Eight baseline records, taken with no event, show where the noise sits. Below 100 kHz it is 11.5 to 26.8 mV rms. Inside the AE band (100 to 500 kHz) five records show only 3.7 to 4.6 mV rms. Events are 234 to 275 kHz. A high-pass just under the band removes the large low-frequency noise before the gain stage amplifies it by 4.9.



2 Derivation from the node equations

U1A is a voltage follower, so the output equals the voltage at its + input: $V_o = V_B$. The current into each node must sum to zero (all voltages are deviations from VMID5, so R2 goes to 0).

Node B (the + input draws no current):

$$(V_B - V_A)\, sC_2 + \frac{V_B}{R_2} = 0 \;\Rightarrow\; V_A = V_B\Big(1 + \frac{1}{sR_2C_2}\Big).$$

Node A:

$$(V_A - V_{in})\, sC_1 + (V_A - V_B)\, sC_2 + \frac{V_A - V_o}{R_1} = 0, \qquad V_o = V_B.$$

Substitute V_A and collect V_B:

$$sC_1V_{in} = V_B\Big[sC_1 + \frac{C_1 + C_2}{R_2C_2} + \frac{1}{sR_1R_2C_2}\Big].$$

Divide by sC_1:

$$\boxed{H(s) = \frac{V_B}{V_{in}} = \frac{s^2}{s^2 + s\,\dfrac{C_1 + C_2}{R_2 C_1 C_2} + \dfrac{1}{R_1 R_2 C_1 C_2}}}$$

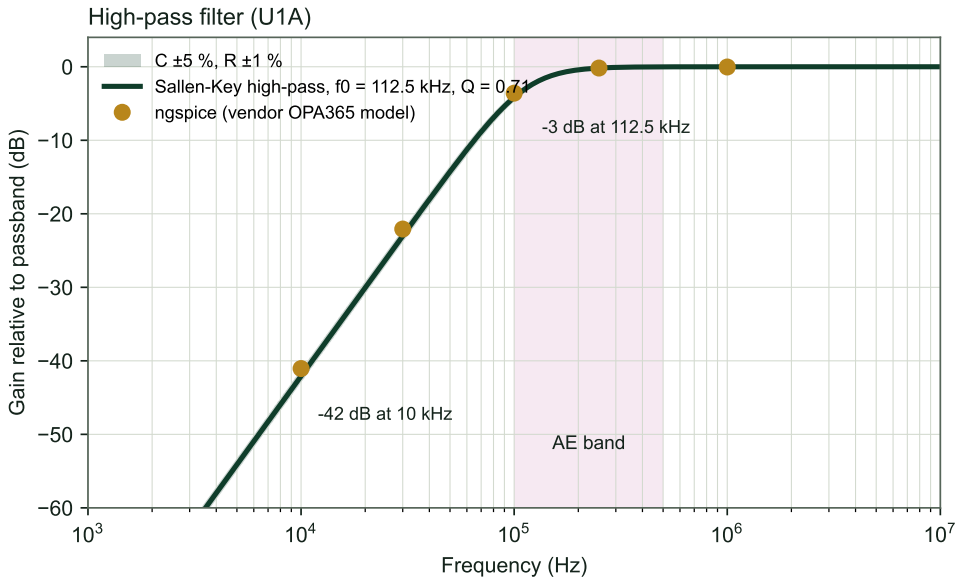
Compare with the standard second-order form $s^2/(s^2 + s\,\omega_0/Q + \omega_0^2)$:

$$\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \qquad \frac{\omega_0}{Q} = \frac{C_1 + C_2}{R_2 C_1 C_2}.$$

With $C_1 = C_2 = C$:

$$f_0 = \frac{1}{2\pi C\sqrt{R_1 R_2}} = \frac{1}{2\pi \cdot 100\,\text{pF} \cdot \sqrt{10\,\text{k}\Omega \cdot 20\,\text{k}\Omega}} = 112.5\,\text{kHz}, \qquad Q = \frac{1}{2}\sqrt{\frac{R_2}{R_1}} = \frac{1}{2}\sqrt{2} = 0.707.$$

$Q = 0.707$ is the Butterworth value: the flattest magnitude with no peak. The rejection below the corner is -40 dB per decade, because the numerator is s^2. At 10 kHz: $(10/112.5)^2 \approx 0.008 = -42$ dB.



3 Why these values

- **Corner at 112.5 kHz**: just under the AE band. A higher corner would also cut the events. The loss at 250 kHz is only 0.17 dB.
- $C_1 = C_2$: then Q depends only on the resistor ratio. Equal C0G capacitors in one value keep the BOM short.
- R_2 **to VMID5, not ground**: the stage rests at 1.65 V, so the single supply never clips the negative half.
- **OPA2365, GBW** 50 MHz: the follower stays accurate far above the corner. Finite gain changes H by less than 0.01 dB at 250 kHz.

4 Tolerances

Capacitors are C0G ±5 % (C14858), resistors ±1 % (C25804, C4184). Over all 16 corner combinations of C_1, C_2, R_1, R_2:

$$f_0 = 106.1 \text{ to } 119.7\,\text{kHz}, \qquad Q = 0.699 \text{ to } 0.714.$$

The corner moves about $\pm 6\,\%$. Q stays within $1\,\%$ of 0.707, so the response keeps its shape. The grey band in the plot is a random sample of these tolerances.